



Modeling the optimal seasonal monoclonal antibody administration strategy for respiratory syncytial virus (RSV) prevention based on age-season specific hospitalization rate of RSV in Suzhou, China, 2016–2022

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ABSTRACT

Background: The approval of nirsevimab brings light to reducing the heavy disease burden caused by respiratory syncytial virus (RSV). Considering the seasonality of RSV, the timing of administering monoclonal antibody (mAb) is critical to maximize health utility. This study aimed to model and seek the optimal seasonal mAb administration strategy for preventing RSV-associated hospitalization.

Methods: Age-season specific hospitalization rates for RSV-associated acute lower respiratory infection (RSV-ALRI) were estimated from a hospital-based birth cohort. Using these rates, we simulated and evaluated the effect of diverse mAb administration strategies on preventing RSV-ALRI hospitalization. Optimal strategies were selected based on their effectiveness and relative cost-effectiveness.

Results: Compared with the year-round strategy of administration mAb at birth for all children, 291 out of the 854 candidate strategies, featuring diverse administration timing and age thresholds, demonstrated a greater number of averted RSV-ALRI hospitalizations and a lower number needed to treat (NNT). The NNT represents the number of mAb doses needed to prevent one case of RSV-ALRI hospitalization. Among the 291 strategies, administration mAb to children born in July–January or August–January at birth and administering to the remaining <12 months old children in September, exhibited the highest increase in averted RSV-ALRI hospitalizations than the year-round strategy, with a magnitude of 23 %, while also achieve an 18 % reduction in NNT.

Conclusion: Administering monoclonal antibodies to children born in July to January at birth, and administering to the remaining <1-year-old children in September or October would be the optimal seasonal mAb administration strategy for children in Suzhou, China.

1. Introduction

Respiratory syncytial virus (RSV) is the most common viral pathogen causing acute lower respiratory infection (ALRI) among young children globally [1]. In 2019, RSV caused an estimated 33 million episodes, 3.6 million hospitalizations, and 26,300 in-hospital deaths worldwide in children aged < 5 years [2]. Recent research has suggested a potential association between respiratory syncytial virus infection during infancy and childhood asthma and recurrent wheezing while establishing a

causal relationship remains inconclusive [3,4]. Despite the uncertainty regarding its long-term impact, given the substantial disease burden associated with RSV in pediatric populations, preventing RSV infection holds a significant public health priority.

The RSV prevention measures currently marketed and in various stages of clinical trials encompass maternal vaccines, pediatric vaccines, and pediatric monoclonal antibodies (mAb) [5]. Preceding the approval of nirsevimab, palivizumab is the only prophylaxis for RSV, but its use was restricted to high-risk population for prevent severe infection [6,7].

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